



Tesla SolarCity offer

Tesla has offered 2.8 billion dollars to acquire SolarCity, the solar panel company.
<http://dgit.in/solartes>



Electronic Persons

Motion passed by the EU classify's robots as 'Electronic Persons' with social security.
<http://dgit.in/robotpeople>

Origins

person to use the initial form of the term algorithm was 12th century English philosopher, Adelard de Bath, who used the term algorismus when translating Al-Khwarizmi's Arabic works.

A major series of achievements in the evolution of algorithms came during the 1800s, the first of which was established by English mathematician George Boole, who also penned The Laws of Thought and established Boolean Algebra. In 1847, Boole unified logic with calculations and formed binary algebra, the basis of computing logic of today. In 1888, Giuseppe Peano established the axiomatization of mathematics (an axiom is an empirical rule established in certain fields, and used without exceptions universally). He used equations with symbols to obtain results. These would later go on to become rules that modern day mathematics and algorithms are based on.

Decades later, algorithms of the present form came into being with Alan Turing's computing machine. Alongside, Alonzo Church's Lambda Calculus became the calculating equivalent of Turing Machines, wherein a variable was bound to specific functions or executable equations to carry out operations. Turing's computer model was based on the notion of a human computer and "states of mind" or environments, in which functions written down in symbols would find application. This was further broken down into basic squares, with each contributing to a specific operation. Each of these basic squares would hold a symbol, and each symbol would be finite in its own state and relativity.

A computer's action would depend upon the tally of logic - each of these symbols acting as bits of data, and the computer's "state of mind" being the condition or purpose of the function being executed. Turing also imposed a limit on the number of symbols being tallied at the same time, and computing more complex logic would require the machine to execute similar, consecutive operations. Each of these operations, hence, would be the simplest unit within the executable 'code'. A code would be the entire compilation of the sets of functions, or



Euclid: The foundation stone for possibly the world's first functional algorithm

the algorithm, based on which a problem would be solved.

Turing's logic of algorithms also laid down the foundation for operations involving variable factors, wherein a subsequent symbol would be altered based on the computation between two preceding symbols. In simpler words, when a function between two variables is executed, the resultant would be the figure that would be used in computing with the next variable, thereby giving rise to progressive algorithms that take into account changing values to compute a code. If need be, these operations would also define the outcome, altering the "state of mind" or parameters of a calculation. "We may now construct a machine to do the work of this computer," Turing deduced in his theory.

Meanwhile, the foundation laid by Alan Turing, and also applicable



Alan Turing: The father of modern cryptography and algorithms

to Church's Lambda Calculus (stated earlier), was worked upon by Church's student and American logician, John Barkley Rosser. Defining an "effective mathematical method", Rosser stated that each step of an algorithm has to be precisely defined and not left to variable outcome chances, with a finite number of steps to obtain the outcome. This, in compliance with the foundation work laid down by Alan Turing, led to the need for a computing machine that would operate and execute commands without the need for human intervention.

This was followed by the establishment of Stephen Kleene's 'Algorithmic Theory' in 1943. Kleene stated, "In setting up a complete algorithmic theory, what we do is to describe a procedure, performable for each set of values of the independent variables, which procedure necessarily terminates and in such manner that from the outcome we can read a definite answer, "yes" or "no," to the question, "is the predicate value true?"

Kleene's theory set up the rule that algorithms of today follow - independent, self-sustaining computational functions that would execute operations within a finite set of instructions. This made computing faster, and with the advent of personal computers from the late 1960s, algorithms have seen improvements. Present applications of algorithms are found in every single moment of our daily life. For instance, on a smartphone, the colour balance of photographs captured by its camera is defined by a set of algorithms, which identifies colours and balances contrast based on a scene. With increasing processing power, algorithms have grown in their complexity and level of computational prowess.

The present generation is gunning for quantum computing and artificial intelligence. Machine learning depends on algorithms to "learn" from usage methods and prepare actions based on personal ways of operations. We are on the verge of time where 'bots' are looking to replace apps of our daily use. Algorithms are behind every nascent stage of technology, having evolved in their ability, while keeping the basis of operations constant. 